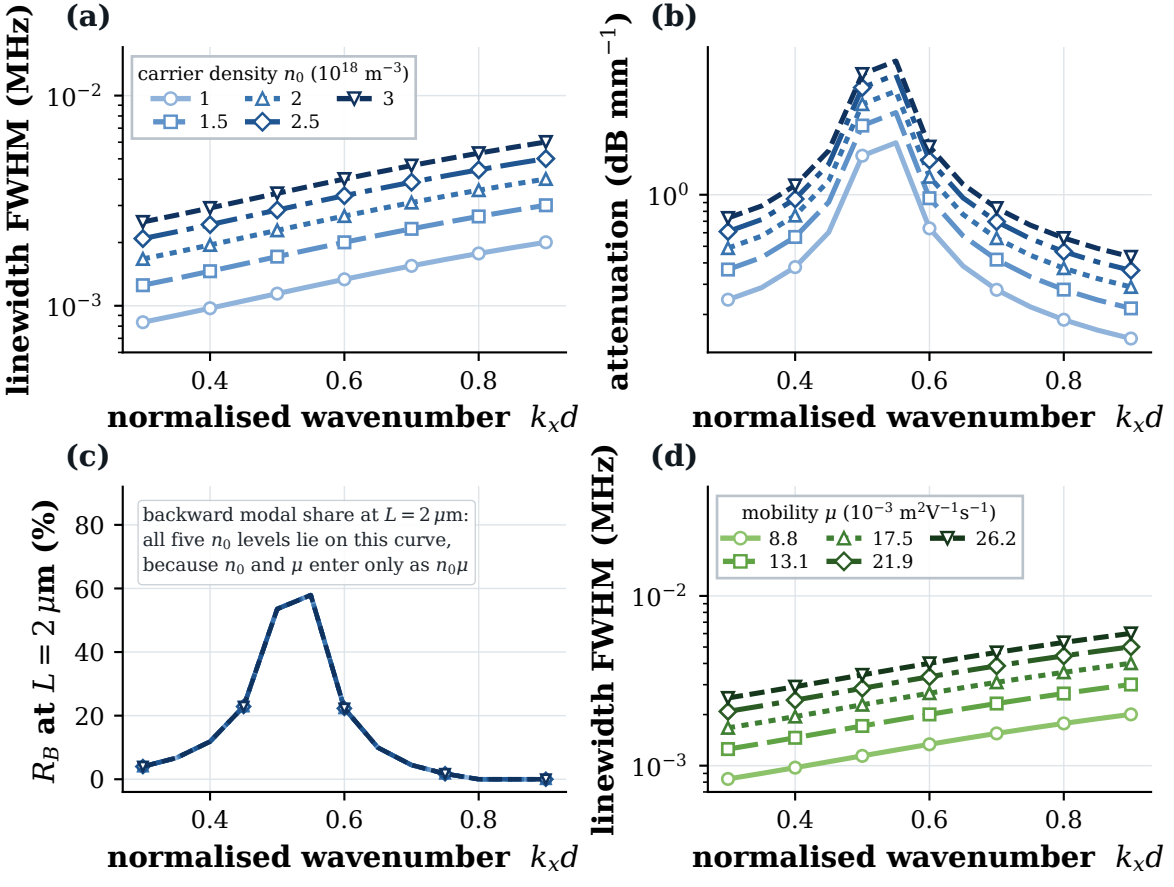
(a)
linewidth FWHM (MHz)
carrier density n_0 (10^{18} m^{-3})
1
1.5
2
2.5
3
10^{-2}
10^{-3}
0.4
0.6
0.8
normalised wavenumber $k_x d$
(b)
attenuation (dB mm^{-1})
10^{0}
0.4
0.6
0.8
normalised wavenumber $k_x d$
(c)
R_B at $L = 2\,\mu$m (%)
backward modal share at $L = 2\,\mu$m:
all five n_0 levels lie on this curve,
because n_0 and μ enter only as $n_0\mu$
80
60
40
20
0
0.4
0.6
0.8
normalised wavenumber $k_x d$
(d)
linewidth FWHM (MHz)
mobility μ (10^{-3} m^{2}V^{-1}s^{-1})
8.8
13.1
17.5
21.9
26.2
10^{-2}
10^{-3}
0.4
0.6
0.8
normalised wavenumber $k_x d$